Technical Characterization Sheet

Institute:	Niels Bohr Institute

Product name:	Single Quantum Eos CS
Product serial number:	SQ222
System CE serial number:	040296
Product description:	12-channel superconducting nanowire single photon detection system

Cold head model:	Sumitomo RDK-101D
Compressor model:	Sumitomo F-20L
Electronic driver:	Single Quantum V4-394 - 4 Channel V4-395 - 8 Channel

Characterization date:	04/04/2024
Characterized by:	Jordy Stad
Approved by:	Jasper van den Hoogen

Cryostat operation

Base temperature (K)	Cooldown time (Hr)
2.5	6

SINGLE QUANTUM
Excellence in photon detection

Specification sheet

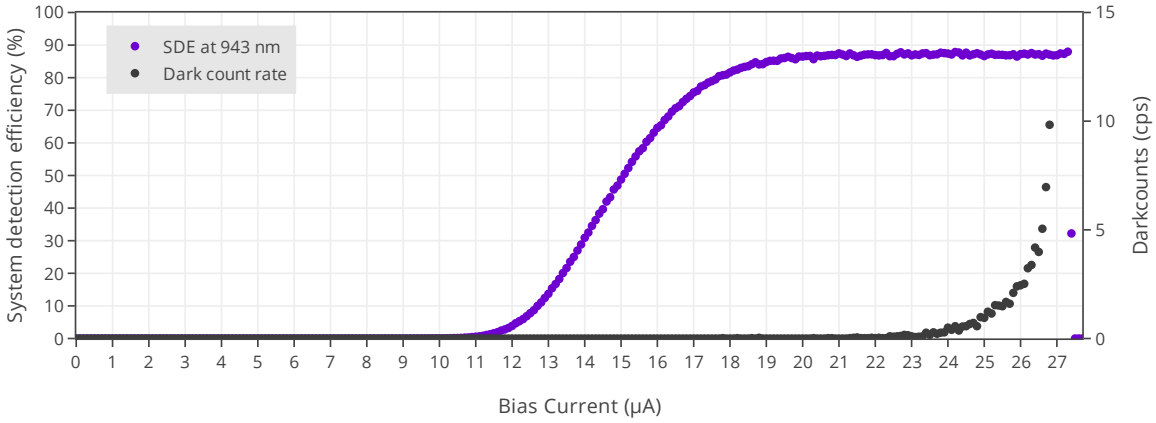
Channel	Driver - channel	SDE at λ nm (%)[1]				DCR (cps)[2]	Timing jitter (ps)[3]	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)
		785	943	1310	1550					
1	V4-394 - 1	-	87	-	-	<5	14	27.7	25.7	1180
2	V4-394 - 2	-	85	-	-	<1	13	30.3	28.4	1230
4	V4-394 - 3	-	81	-	-	<2	12	32.4	30.4	1300
5	V4-394 - 4	-	87	-	-	<2	12	32.4	30.2	1360
7	V4-395 - 7	87	-	-	-	<1	14	32.7	30.9	1390
8	V4-395 - 8	86	-	-	-	<1	14	30.8	28.6	1260
10	V4-395 - 1	-	91	-	-	<2	24	25.0	23.3	1710
12	V4-395 - 2	-	92	-	-	<2	25	25.8	24.0	1680
14	V4-395 - 3	-	91	-	-	<1	19	28.0	25.5	1760
16	V4-395 - 4	-	91	-	-	<1	21	26.5	24.4	1760
17	V4-395 - 5	-	-	76	80	<5	32	24.9	23.6	1370
19	V4-395 - 6	-	-	75	82	<20	42	14.8	13.4	990

[1] System detection efficiency at operating current. Measurement uncertainty ± 3%.
[2] On the next pages, the SDE and DCR are plotted as a function of the bias current.
[3] FWHM value, measurement uncertainty: ± 2 ps. Jitter can be lowered at the expense of higher DCR, by increasing the bias current. More information is given in the next pages.

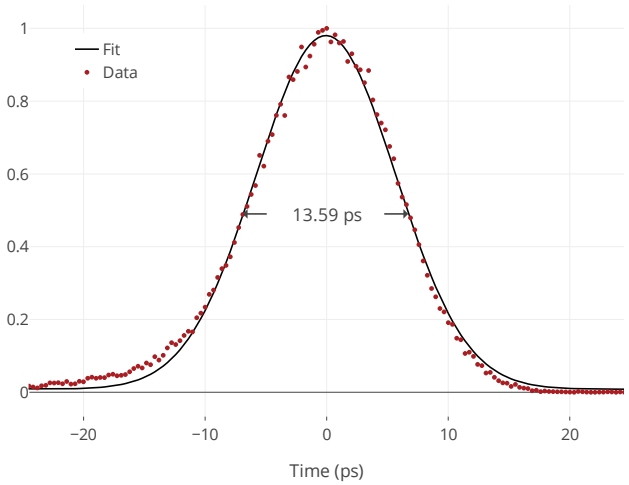
Channel 1

Fiber type: SM780
Driver V4-394: Channel 1

SDE at 943 nm (%)	DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)
87	<5	14	27.7	25.7	1180



SDE and DCR as a function of the bias current



Jitter distribution

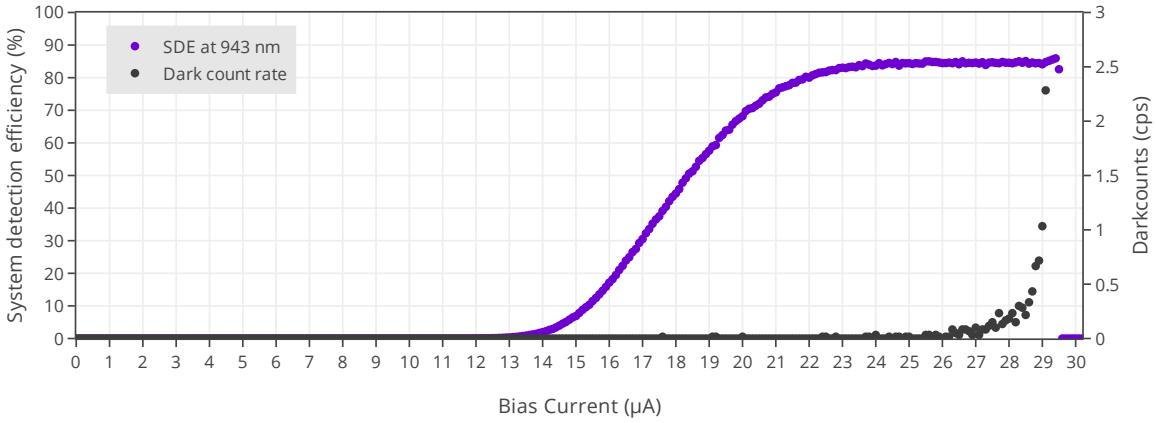
Detector count rate (Mcps)	<0.5	5	10
Bias current (µA)	25.7	22.6	20.2
SDE at 943 nm (%)	87	77	70

High count rate measured with a CW laser.

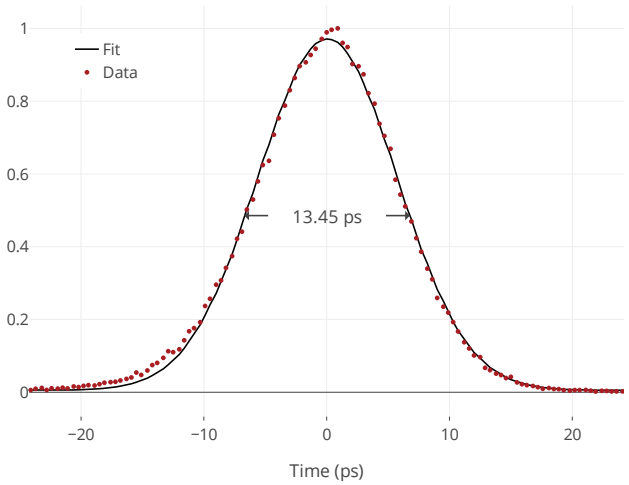
Channel 2

Fiber type: SM780
Driver V4-394: Channel 2

SDE at 943 nm (%)	DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)
85	<1	13	30.3	28.4	1230



SDE and DCR as a function of the bias current



Jitter distribution

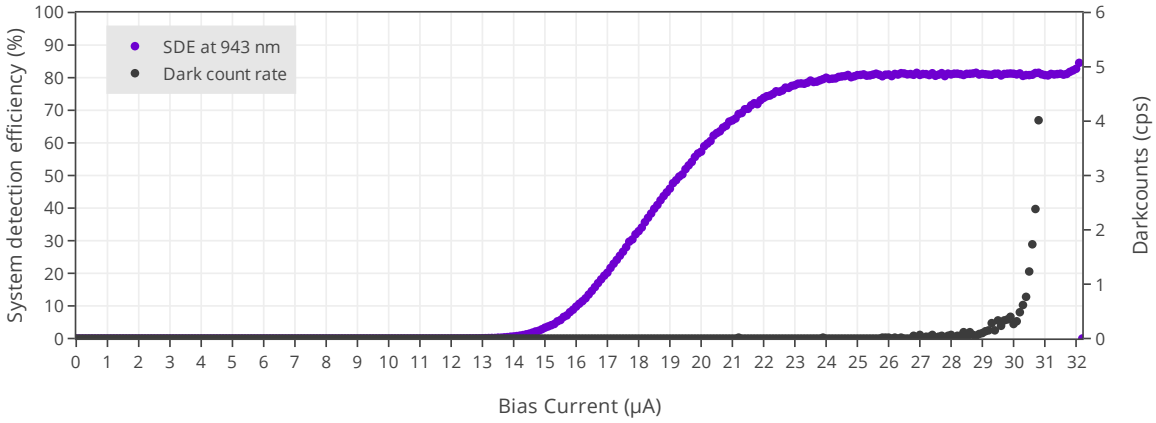
Detector count rate (Mcps)	<0.5	5	10
Bias current (µA)	28.4	25.8	23.5
SDE at 943 nm (%)	85	80	72

High count rate measured with a CW laser.

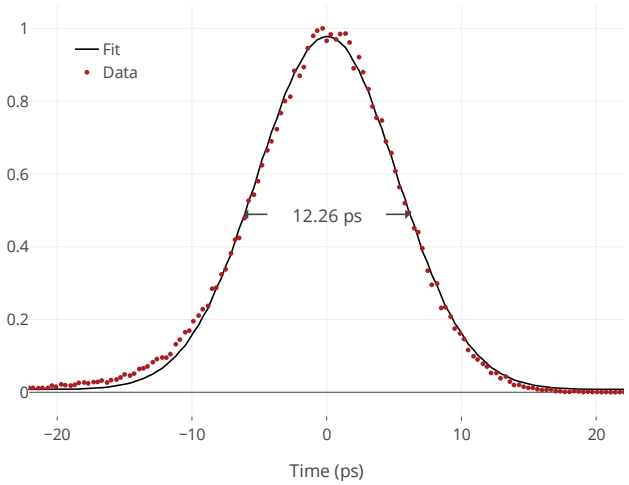
Channel 4

Fiber type: SM780
Driver V4-394: Channel 3

SDE at 943 nm (%)	DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)
81	<2	12	32.4	30.4	1300



SDE and DCR as a function of the bias current



Jitter distribution

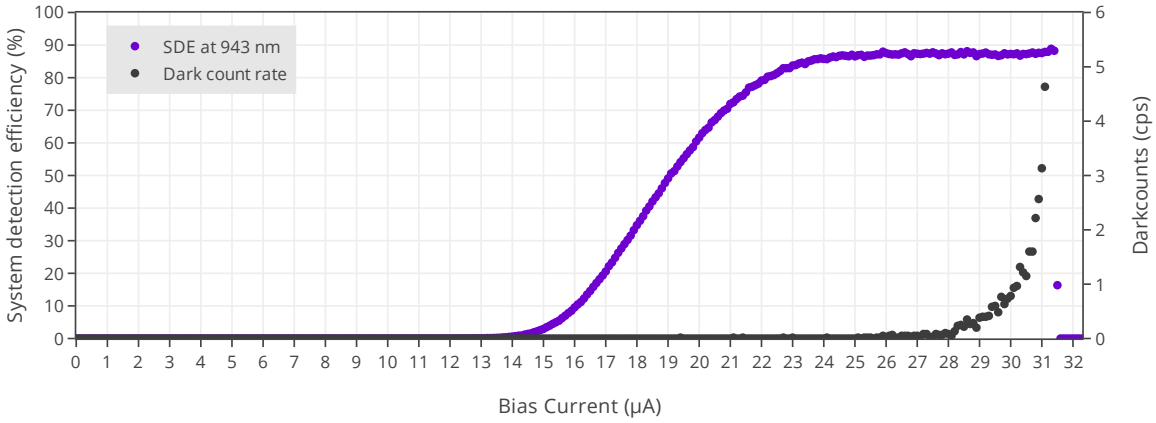
Detector count rate (Mcps)	<0.5	5	10
Bias current (µA)	30.4	28.0	25.5
SDE at 943 nm (%)	81	73	66

High count rate measured with a CW laser.

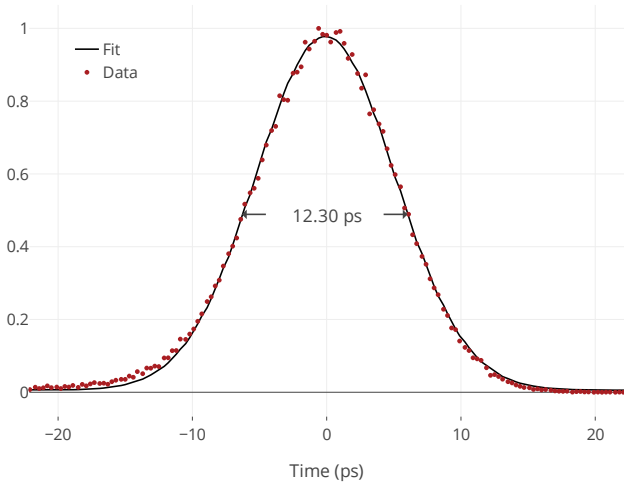
Channel 5

Fiber type: SM780
Driver V4-394: Channel 4

SDE at 943 nm (%)	DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)
87	<2	12	32.4	30.2	1360



SDE and DCR as a function of the bias current



Jitter distribution

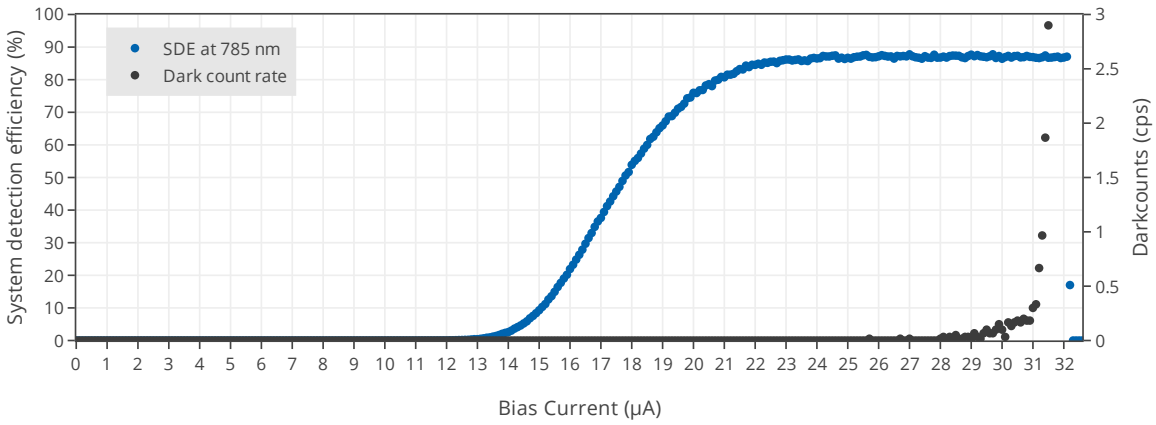
Detector count rate (Mcps)	<0.5	5	10
Bias current (µA)	30.2	26.2	24.3
SDE at 943 nm (%)	87	74	67

High count rate measured with a CW laser.

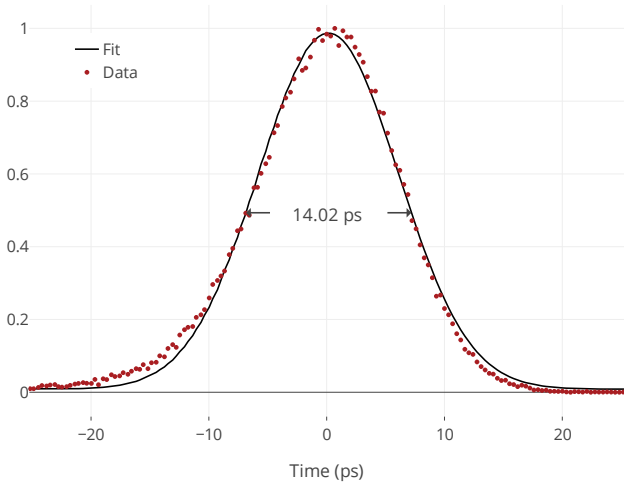
Channel 7

Fiber type: SM780
Driver V4-395: Channel 7

SDE at 785 nm (%)	DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)
87	<1	14	32.7	30.9	1390



SDE and DCR as a function of the bias current



Jitter distribution

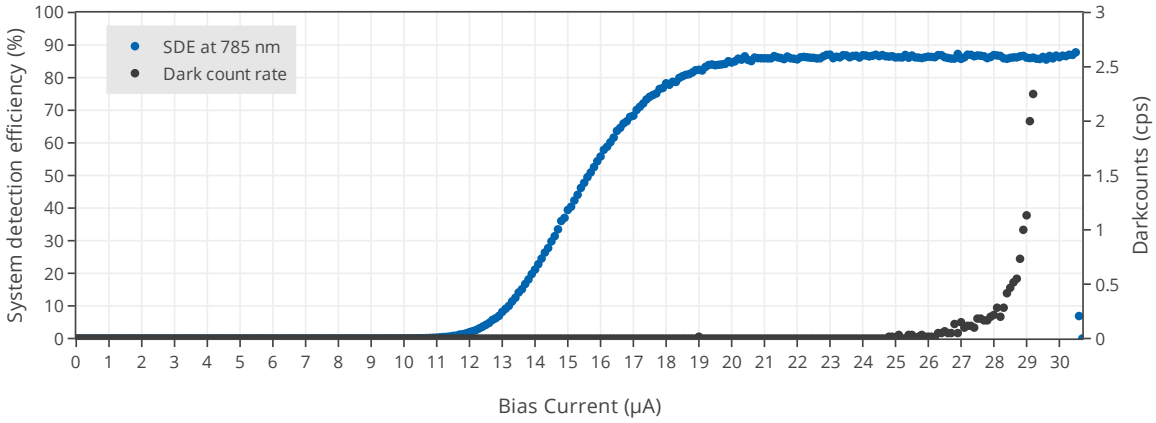
Detector count rate (Mcps)	<0.5	5	10	20
Bias current (µA)	30.9	25.7	23.2	19.5
SDE at 785 nm (%)	87	72	64	51

High count rate measured with a CW laser.

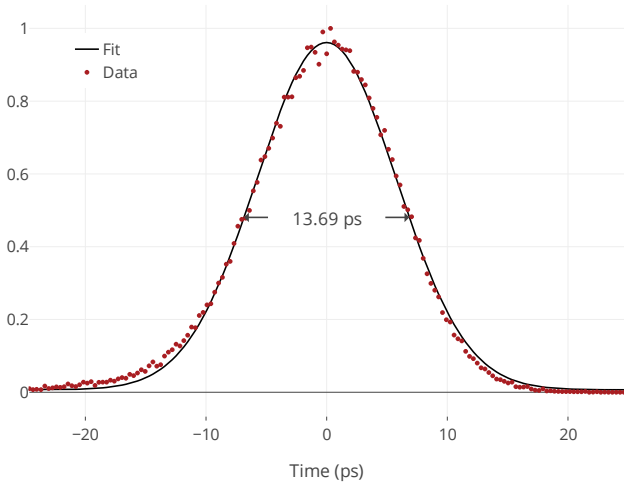
Channel 8

Fiber type: SM780
Driver V4-395: Channel 8

SDE at 785 nm (%)	DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)
86	<1	14	30.8	28.6	1260



SDE and DCR as a function of the bias current



Jitter distribution

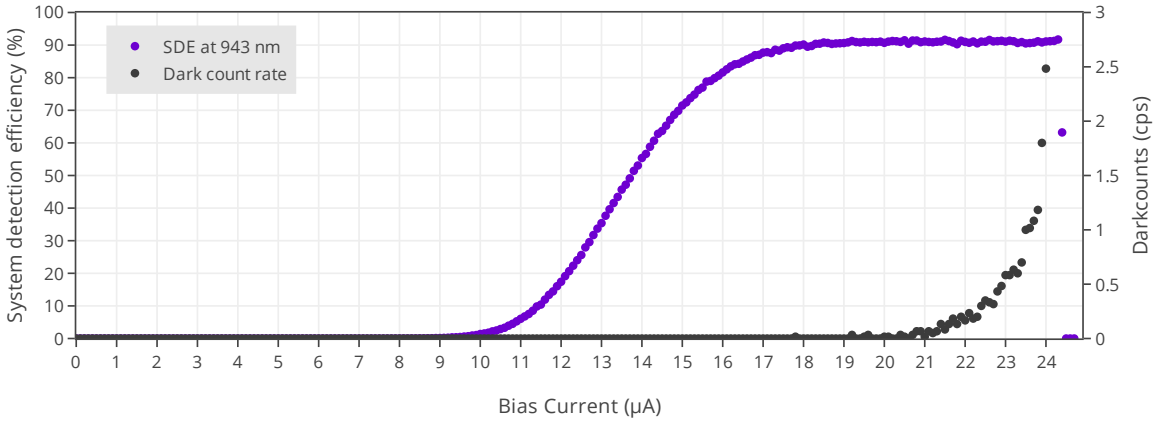
Detector count rate (Mcps)	<0.5	5	10	20
Bias current (µA)	28.6	25.1	23.5	19.3
SDE at 785 nm (%)	86	76	70	62

High count rate measured with a CW laser.

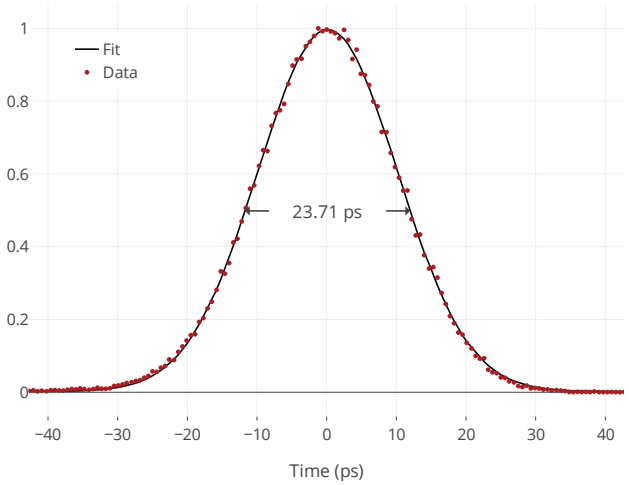
Channel 10

Fiber type: SM780
Driver V4-395: Channel 1

SDE at 943 nm (%)	DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)
91	<2	24	25.0	23.3	1710



SDE and DCR as a function of the bias current



Jitter distribution

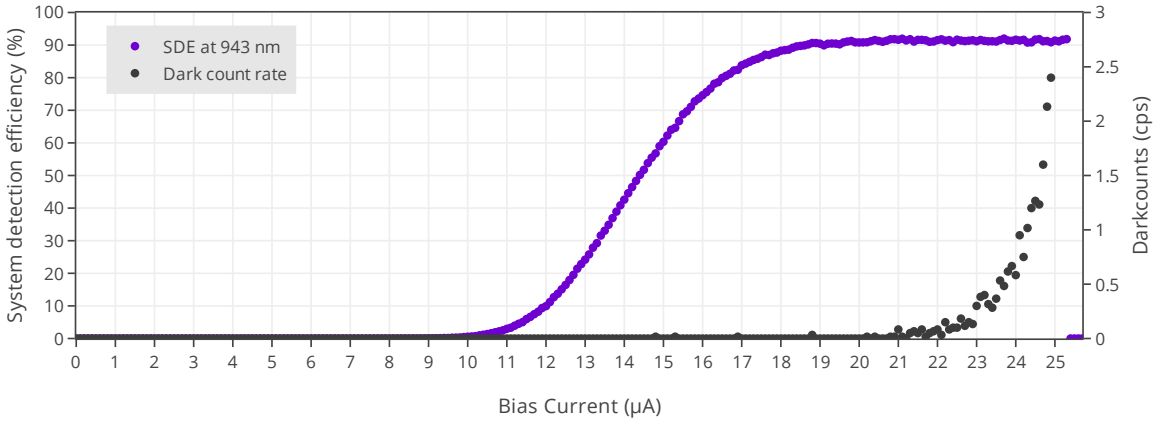
Detector count rate (Mcps)	<0.5	5	10
Bias current (µA)	23.3	19.6	17.6
SDE at 943 nm (%)	91	76	66

High count rate measured with a CW laser.

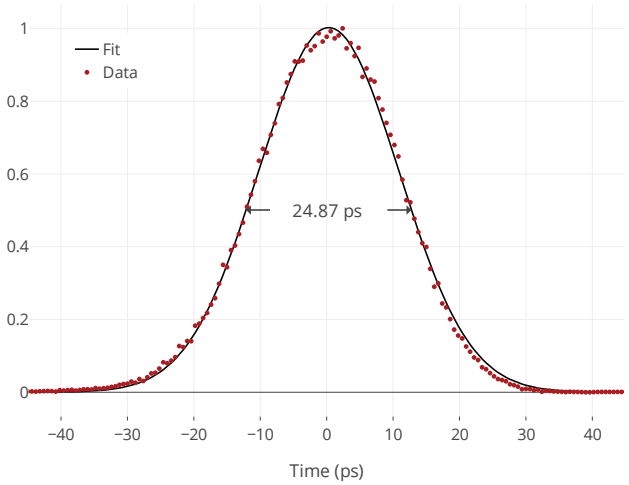
Channel 12

Fiber type: SM780
Driver V4-395: Channel 2

SDE at 943 nm (%)	DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)
92	<2	25	25.8	24.0	1680



SDE and DCR as a function of the bias current



Jitter distribution

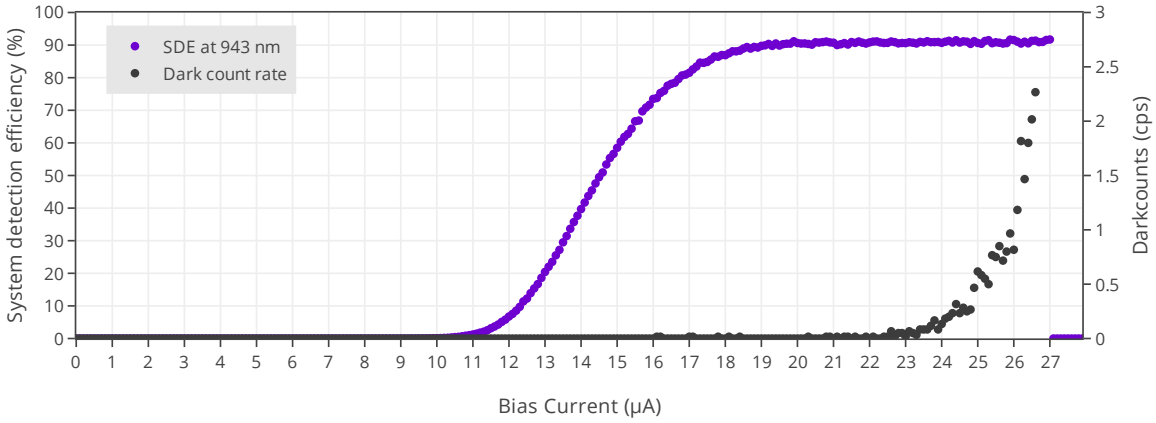
Detector count rate (Mcps)	<0.5	5	10
Bias current (µA)	24.0	20.1	18.2
SDE at 943 nm (%)	92	79	68

High count rate measured with a CW laser.

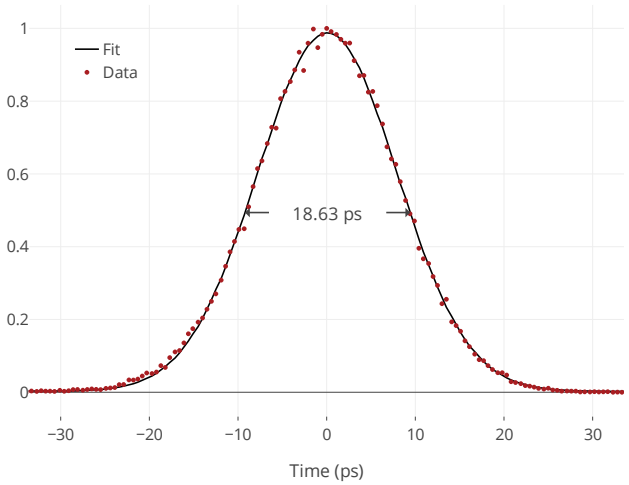
Channel 14

Fiber type: SM780
Driver V4-395: Channel 3

SDE at 943 nm (%)	DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)
91	<1	19	28.0	25.5	1760



SDE and DCR as a function of the bias current



Jitter distribution

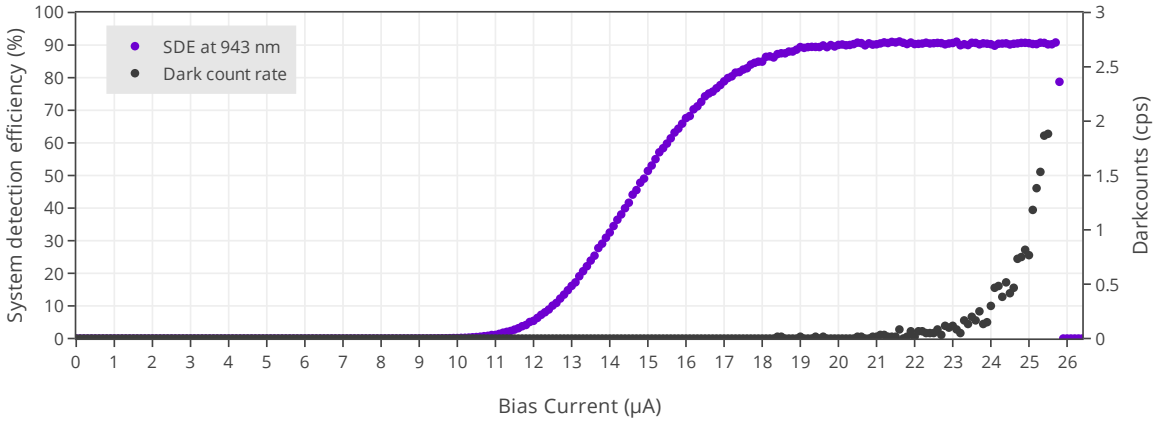
Detector count rate (Mcps)	<0.5	5	10
Bias current (µA)	25.5	22.4	20.0
SDE at 943 nm (%)	91	78	68

High count rate measured with a CW laser.

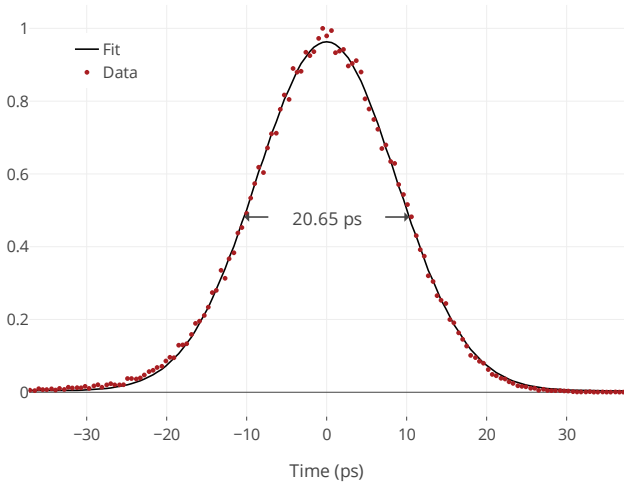
Channel 16

Fiber type: SM780
Driver V4-395: Channel 4

SDE at 943 nm (%)	DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)
91	<1	21	26.5	24.4	1760



SDE and DCR as a function of the bias current



Jitter distribution

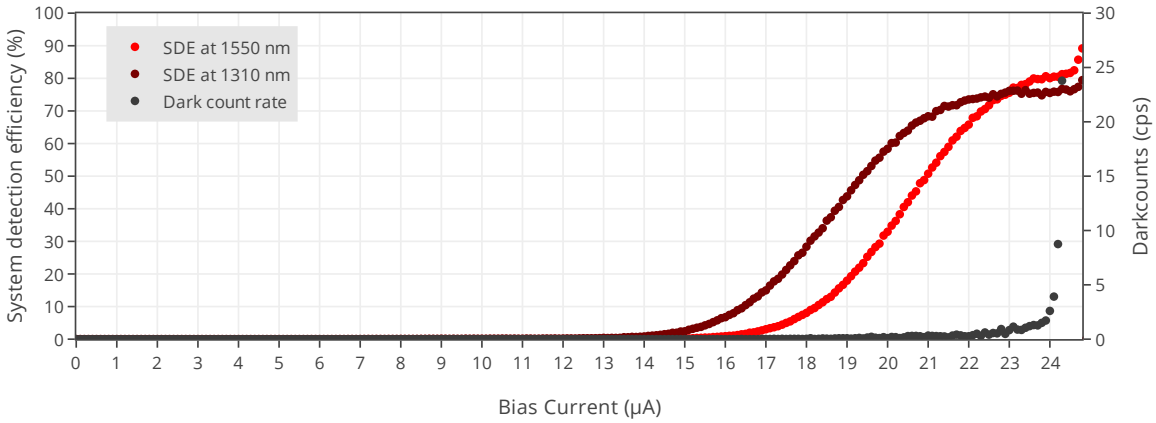
Detector count rate (Mcps)	<0.5	5	10
Bias current (µA)	24.4	21.2	19.0
SDE at 943 nm (%)	91	76	66

High count rate measured with a CW laser.

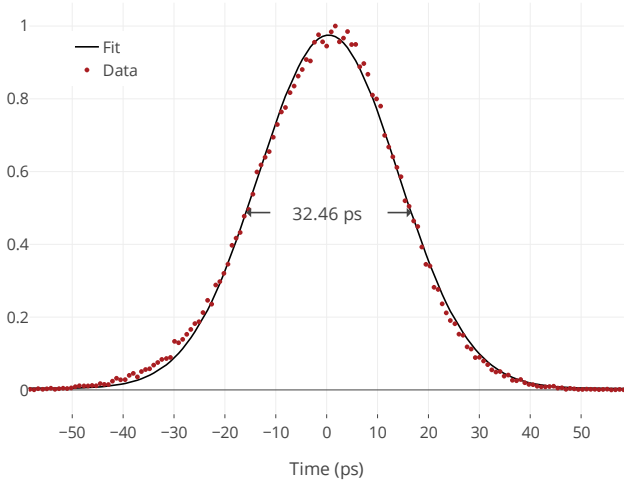
Channel 17

Fiber type: SMF28
Driver V4-395: Channel 5

SDE at λ nm (%)		DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)
1310	1550					
76	80	<5	32	24.9	23.6	1370



SDE and DCR as a function of the bias current



Jitter distribution

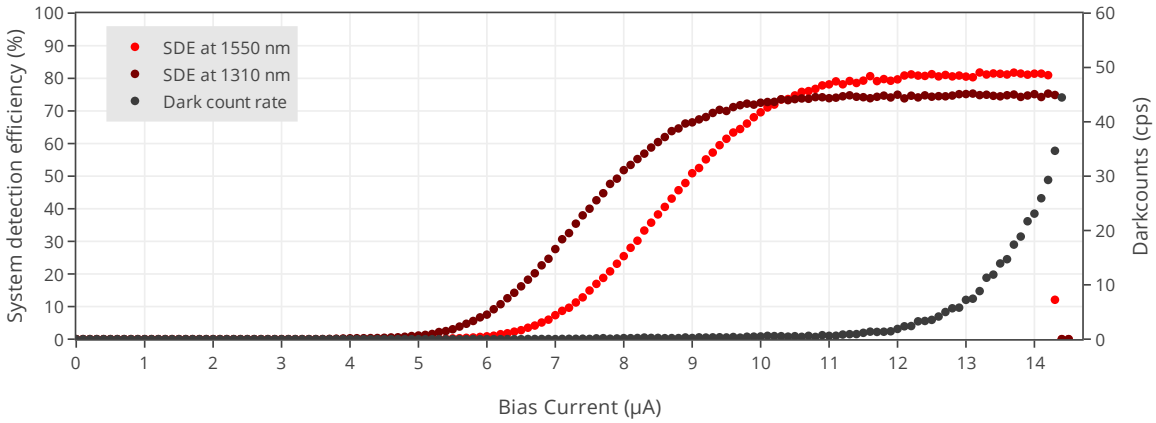
Detector count rate (Mcps)	<0.5	4.0	6.2	8.8	10.0
Bias current (µA)	23.6	22.7	21.6	21.5	21.7
SDE at 1550 nm (%)	80	80	80	80	80

High count rate measured with a pulsed laser with a repetition rate of 11 MHz

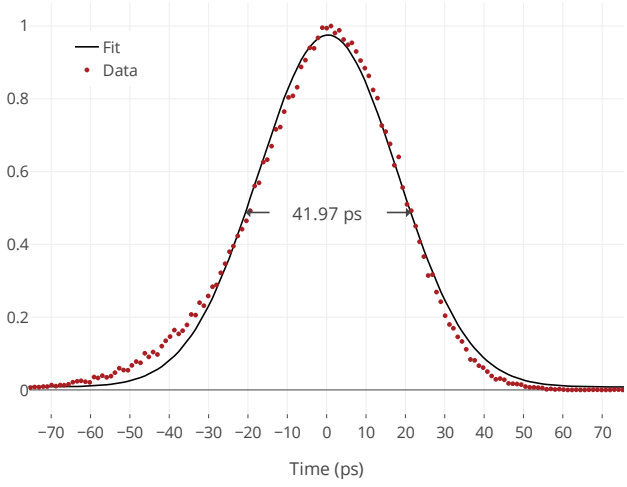
Channel 19

Fiber type: SMF28
Driver V4-395: Channel 6

SDE at λ nm (%)		DCR (cps)	Timing jitter (ps)	Critical current (µA)	Operating current (µA)	Output pulse amplitude (mV)
1310	1550					
75	82	<20	42	14.8	13.4	990



SDE and DCR as a function of the bias current



Jitter distribution

Detector count rate (Mcps)	<0.5	3.8	5.8	8.9	10.0
Bias current (µA)	13.4	10.0	8.7	8.2	7.9
SDE at 1550 nm (%)	82	78	76	76	76

High count rate measured with a pulsed laser with a repetition rate of 11 MHz

Appendix 1: Characterization methods

Efficiency

The efficiency curve is generated by coupling either a 785 nm or 943 nm laser source at 10 nW, or a 1310 nm or 1550 nm laser source at 1.5 nW, which will be attenuated with 50 dB. This can be calculated as 395k, 475k, 99k or 117k photons coming in every second, respectively. With the photon source attached we sweep the bias current over the detectors with a step size of 0.1 µA and an integration time of 200 ms.

With the known input photon flux, the system detection efficiency (SDE) can be calculated by: measured count rate / input photon flux x100%

Dark count rate

The dark count rate is measured by sweeping the bias current over the detectors at 0.1 µA intervals, with an integration time of 60 s. During this measurement, no optical fiber is attached to the system.

Timing Jitter

The jitter is measured with a fast pulse laser with a built-in fast photodiode (Fianium FM-1060) and a high-end oscilloscope. Laser pulses of <5 ps are generated with a repetition rate of 50 MHz. This signal is optically coupled to the SNSPD. The output of both the built-in photodiode and the SNSPD are coupled into the oscilloscope. The oscilloscope is set to trigger at the steepest part of the detection pulse of the SNSPD.

The oscilloscope measures the time difference between the electrical output of the laser source and the electrical output of the SNSPD and stores the measured data in a histogram.

From this distribution, the jitter can be calculated by measuring the full width at half maximum (FWHM).

Efficiency at high count rates

With increasing input photon flux the chance of a photon not being detected due to it arriving at the SNSPD within the recovery time of the SNSPD increases, decreasing detection efficiency. We therefore measure the SDE at increasing count rates. We use a pulsed and CW laser at a fixed repetition rate for the pulsed laser and increase the photon flux in steps. We use this to measure the detection efficiency as a function of detector count-rate.